TEACHER REFERENCE · 35-QUESTION KEY

Final Exam

Complete answer key with worked solutions for every multiple-choice item and free-response sub-part. Use for grading the Canvas-submitted student responses, or release to students after the exam window closes.

4 Units | 28 MC + 7 FRQ Sub-Parts | Worked Solutions | Common-Error Notes

QUICK REFERENCE

Multiple Choice · Q1–Q28

Q01 E	Q02 C	Q03 C	Q04 D	Q05 F	Q06 B	Q07 E
Q08 D	Q09 F	Q10 A	Q11 D	Q12 B	Q13 E	Q14 F
Q15 B	Q16 D	Q17 A	Q18 D	Q19 E	Q20 A	Q21 C
Q22 A	Q23 C	Q24 E	Q25 F	Q26 B	Q27 F	Q28 B

UNIT 01 Polynomial & Rational Functions

01

1A · TOPIC 1.2 AVG RATE OF CHANGE

ANSWER E

A function f satisfies f(2) = 5 and f(7) = 20. What is the average rate of change of f on the interval [2, 7]?

A -3

B $\frac{1}{3}$

C 5

D $\frac{25}{9}$

E 3

F 15

WORKED SOLUTION

AROC on [a, b] = (f(b) − f(a)) / (b − a). Plug in: (f(7) − f(2)) / (7 − 2) = (20 − 5) / 5 = 15 / 5 = 3. COMMON ERROR: stopping at 15 (forgetting to divide by Δx).

02

1A · TOPIC 1.3 CONCAVITY

ANSWER C

A continuous function f is increasing on [0, 10]. On the interval (0, 5), f is concave down; on the interval (5, 10), f is concave up. What feature does the graph have at x = 5?

A Local maximum

B Local minimum

C Point of inflection

D Vertical asymptote

E Horizontal asymptote

F Removable discontinuity

WORKED SOLUTION

f is increasing on [0, 10], so the function never changes direction — there's no max or min. Concavity flips from concave-down (∩) to concave-up (∪) at x = 5. **Concavity flip with no direction flip = point of inflection.**

03

1B · TOPIC 1.4 🔥 BIGGEST MISS

ANSWER C

The graph of a polynomial f is increasing on the entire interval (1, 7). Between x = 3 and x = 5 the curve has an obvious S-bend where the concavity switches from concave up to concave down at $x = 4$. The point at x = 4 is best classified as:

A Local maximum (the curve reaches a peak there)

B Local minimum (the curve reaches a valley there)

C Point of inflection (concavity changes; direction does not)

D Global maximum on the interval

E Both a local max and a point of inflection

F A vertical asymptote

WORKED SOLUTION

THIS IS THE EXACT MISCONCEPTION YOUR CLASS MISSED ON PRACTICE EXAM Q2. f is increasing throughout (1, 7), so the curve never reaches a peak or valley —

extrema are impossible. Concavity changing alone (concave-up → concave-down) means a **point of inflection**, not a max.

04

1B · TOPIC 1.6 END BEHAVIOR

ANSWER D

For the polynomial $f(x) = -2x^4 + 5x^3 - x + 9$, which statement correctly describes the end behavior?

A As $x \to \pm\infty$, $f(x) \to +\infty$

B As $x \to -\infty$, $f(x) \to -\infty$ and as $x \to +\infty$, $f(x) \to +\infty$

C As $x \to -\infty$, $f(x) \to +\infty$ and as $x \to +\infty$, $f(x) \to -\infty$

D As $x \to \pm\infty$, $f(x) \to -\infty$

E As $x \to \pm\infty$, $f(x) \to 0$

F End behavior cannot be determined without more information

WORKED SOLUTION

End behavior is determined by the leading term. Even degree (4) with negative leading coefficient (−2) ⇒ both ends point down. $x \to -\infty \Rightarrow f(x) \to -\infty$ AND $x \to +\infty \Rightarrow f(x) \to -\infty$.

05

1B · TOPIC 1.5 MULTIPLICITY

ANSWER F

For the polynomial $f(x) = 3(x + 2)^2(x - 1)(x - 4)^3$, identify the behavior of the graph at each x-intercept.

A $x = -2$ S-crosses, $x = 1$ bounces, $x = 4$ crosses cleanly

B $x = -2$ crosses cleanly, $x = 1$ bounces, $x = 4$ S-crosses

C All three intercepts cross cleanly

D $x = -2$ bounces, $x = 1$ S-crosses, $x = 4$ crosses cleanly

E $x = -2$ bounces, $x = 1$ crosses cleanly, $x = 4$ bounces

F $x = -2$ bounces, $x = 1$ crosses cleanly, $x = 4$ S-crosses

WORKED SOLUTION

Multiplicity dictates behavior at each zero of $f(x) = 3(x + 2)^2(x - 1)(x - 4)^3$: **$x = -2$ (mult 2) bounces**, **$x = 1$ (mult 1) crosses cleanly**, **$x = 4$ (mult 3) S-curve crosses**. Even multiplicity = bounce; odd ≥ 3 = S-shape cross.

06 1C · TOPIC 1.7 HOLES VS ASYMPTOTES ANSWER B

For the rational function $r(x) = (x - 3)(x + 1) / [(x - 3)(x + 5)]$, identify all holes and vertical asymptotes.

A Vertical asymptotes at $x = 3$ and $x = -5$; hole at $x = -1$

B Vertical asymptote at $x = -5$; hole at $x = 3$

C Holes at $x = 3$ and at $x = -5$

D Vertical asymptotes at $x = 3$ and $x = -5$ (no holes)

E Vertical asymptote at $x = 3$; hole at $x = -5$

F No holes; vertical asymptote at $x = -1$ only

WORKED SOLUTION

Factor: numerator $(x - 3)(x + 2)$, denominator $(x + 5)(x - 3)$. $x = 3$ zeroes both top and bottom ⇒ **removable hole** at $x = 3$. $x = -5$ zeroes only the denominator ⇒ **vertical asymptote** at $x = -5$.

07

1C · TOPIC 1.8 ASYMPTOTES

ANSWER E

Which statement best describes the long-run behavior of $f(x) = (4x^3 + 1) / (x^2 + 5)$?

A Horizontal asymptote at $y = 0$

B Horizontal asymptote at $y = 4$

C Horizontal asymptote at $y = 1/5$

D Horizontal asymptote at $y = 4x$

E No horizontal asymptote — the function has a slant (oblique) asymptote

F Horizontal asymptote at $y = \infty$

WORKED SOLUTION

Compare degrees of numerator and denominator. Numerator has degree 2, denominator has degree 1. Since deg(num) > deg(den), there is **no horizontal asymptote**. Specifically (deg num) − (deg den) = 1, so there's a **slant (oblique) asymptote** instead.

UNIT 02 Exp / Log / Inverse

08

2A · TOPIC 2.2 GEOMETRIC SEQ

ANSWER D

A geometric sequence has first term $a_1 = 6$ and common ratio $r = 2$. What is the value of the 6th term, a_6?

A 12

B 36

C 96

D 192

E 384

F 64

WORKED SOLUTION

Geometric formula: $a_n = a_1 \cdot r^{(n-1)}$. With $a_1 = 6$, $r = 2$, $n = 6$: $a_6 = 6 \cdot 2^5 = 6 \cdot 32 = 192$. COMMON ERROR: using r^n instead of $r^{(n-1)}$ gives $6 \cdot 2^6 = 384$ (option E).

09 2A · TOPIC 2.4 FRACTIONAL EXPONENTS ANSWER F

Simplify the expression $27^{(2/3)}$.

A 18

B 6

C 81

D $\frac{27}{2}$

E 3

F 9

WORKED SOLUTION

Fractional exponent rule: take the root first, then the power. $27^{(1/3)} = 3$, then $3^2 = 9$. COMMON ERRORS: 18 (treating 2/3 as multiplication: $27 \cdot 2/3$) or 27/2 (treating as a ratio).

10

2B · TOPIC 2.12 LOG PROPERTIES

ANSWER A

Which expression is equivalent to $\log(x^2\sqrt{y} / z^3)$, where the $\sqrt{}$ denotes square root and all variables are positive?

- **A** $2 \log x + (1/2) \log y - 3 \log z$
- **B** $2 \log x \cdot (1/2) \log y / 3 \log z$
- **C** $\log(2x) + \log(\sqrt{y}) - \log(3z)$
- **D** $2 \log x + \log(\frac{1}{2}) \cdot \log y - 3 \log z$
- **E** $2 \log x + (1/2) \log y + 3 \log z$
- **F** $\log(2x + \sqrt{y} - 3z)$

WORKED SOLUTION

Apply the three log rules: product ($\times \to +$), quotient ($\div \to -$), power (exponent → coefficient). Note $\sqrt{y} = y^{(1/2)}$. $\log(x^2\sqrt{y} / z^3) = \log(x^2) + \log(y^{(1/2)}) - \log(z^3) = 2 \log x + (1/2) \log y - 3 \log z$.

11

2B · TOPIC 2.13 LOG EQUATION

ANSWER D

Solve for x: $\log_4(x) + \log_4(x - 6) = 2$.

- **A** $x = 2$
- **B** $x = 4$
- **C** $x = 6$
- **D** $x = 8$
- **E** $x = 16$

F $x = 22$

WORKED SOLUTION

Convert log to exponential: $\log_2(x) = 3$ means $2^3 = x$, so $x = 8$. COMMON ERROR YOUR CLASS MADE (PRACTICE EXAM Q15): computing $2 \cdot 3 = 6$ instead of 2^3.

12

2B · TOPIC 2.13 🔥 BIGGEST MISS · 5/10 WRONG

ANSWER B

What is the x-intercept of the function $f(x) = \log_3(x - 1) - 2$?

A $x = 1$

B $x = 10$

C $x = 4$

D $x = 9$

E $x = 2$

F $x = 27$

WORKED SOLUTION

BIGGEST MISS · 5/10 WRONG. Set $f(x) = 0$: $\log_3(x - 1) - 2 = 0 \Rightarrow \log_3(x - 1) = 2 \Rightarrow x - 1 = 3^2 = 9 \Rightarrow x = 10$. COMMON ERROR: stopping at $x - 1 = 9$ and answering $x = 9$ (option D), forgetting to add 1 back.

13

2B · TOPIC 2.15 🔥 SEMI-LOG · 4/10 WRONG

ANSWER E

When the points (x, y) of a data set are graphed as $(x, \log_{10} y)$ on a semi-log plot, the result is a line with slope 0.5 and y-intercept 1. Which formula best models the original function $y(x)$?

A $y = 0.5x + 1$ (simple linear)

B $y = \log(0.5x + 1)$

C $y = 0.5 \cdot 10^x$

D $y = x^{0.5} + 10$

E $y = 10 \cdot 10^{0.5x}$

F $y = e^{0.5x + 1}$

WORKED SOLUTION

On a semi-log plot (log y vs. x), exponential data $y = a \cdot b^x$ becomes a straight line. If the line of best fit is $\log y = 1 + 0.5x$, then $y = 10^{1 + 0.5x} = 10^1 \cdot 10^{0.5x} = 10 \cdot 10^{0.5x}$.

14

2C · TOPIC 2.8 🔥 HIGHEST MISS IN UNIT 2

ANSWER F

The graph of f passes through the points (1, 4), (3, 7), and (5, 12). Through which set of points must the graph of f^{-1} (the inverse of f) pass?

A (1, ¼), (3, 1/7), (5, 1/12)

B (-1, -4), (-3, -7), (-5, -12)

C (1, 4), (3, 7), (5, 12)

D (4, -1), (7, -3), (12, -5)

E (¼, 1), (1/7, 3), (1/12, 5)

F (4, 1), (7, 3), (12, 5)

WORKED SOLUTION

Parametric circle: $x = 2 + 2\cos(t)$, $y = 1 + 2\sin(t)$. Evaluate at the listed t-values: $t = 0 \Rightarrow (2 + 2, 1 + 0) = (4, 1)$; $t = \pi/2 \Rightarrow (2 + 0, 1 + 2) = (2, 3)$; $t = \pi \Rightarrow (0, 1)$; $t = 3\pi/2 \Rightarrow (2, -1)$. Match against the option triples ⇒ (4, 1), (?, 3), (?, ?) appears in F.

UNIT 03 Trigonometry & Polar

15

3A · TOPIC 3.2 🔥 RADIAN DEF · 5/10 WRONG

ANSWER B

Which statement is the correct **definition** of one radian?

A An angle equal to $180/\pi$ degrees exactly

B The central angle of a circle that subtends an arc whose length equals the radius of the circle

C An angle whose terminal side passes through the point (1, 0)

D An angle of measure $360°/\pi$

E Half of one full revolution around a circle

F The angle whose sine equals 1

WORKED SOLUTION

THIS WAS PRACTICE EXAM 3 Q16 — 5/10 WRONG. Definition of a radian: **the central angle whose intercepted arc length equals the radius.** Equivalently: 1 rad = the angle for which (arc length s) = (radius r). All other options describe degree measures or unrelated definitions.

16

3A · TOPIC 3.4 EXACT VALUES

ANSWER D

Find the exact value of $\cos(7\pi/6)$.

A $\frac{1}{2}$

B $-\frac{1}{2}$

C $\frac{\sqrt{3}}{2}$

D $-\frac{\sqrt{3}}{2}$

E $\frac{\sqrt{2}}{2}$

F $-\frac{\sqrt{2}}{2}$

WORKED SOLUTION

$5\pi/3$ lies in Quadrant IV with reference angle $\pi/3$. In QIV: cos is positive, sin is negative. $\sin(\pi/3) = \sqrt{3}/2$, so $\sin(5\pi/3) = -\sqrt{3}/2$.

17 3B · TOPIC 3.6 A·SIN(BX+C)+D ANSWER A

A sinusoidal function is given by $f(x) = 3\cdot\sin(\pi x/4) + 5$. What is its period?

A 8

B 4

C $\pi/4$

D 2π

E 8π

F π

WORKED SOLUTION

For $y = A \sin(Bx) + D$, the period is $2\pi / |B|$. Here $B = \pi/4$, so period $= 2\pi / (\pi/4) = 2\pi \cdot 4/\pi = 8$.

18

3B · TOPIC 3.7 SINUSOIDAL MODELING

ANSWER D

A Ferris wheel has a diameter of 60 m and its center is 35 m above the ground. It completes one full revolution every 4 minutes. A rider boards at the lowest point at time $t = 0$. Which equation models the rider's height $h(t)$ above the ground (in meters)?

A $h(t) = 30 \cdot \sin(\pi t/2) + 35$

B $h(t) = 60 \cdot \cos(\pi t/4) + 35$

C $h(t) = 30 \cdot \cos(\pi t/2) + 35$

D $h(t) = -30 \cdot \cos(\pi t/2) + 35$

E $h(t) = 35 \cdot \sin(\pi t/2) + 30$

F $h(t) = -60 \cdot \sin(\pi t/4) + 35$

WORKED SOLUTION

Ferris wheel modeling. Diameter 60 ⇒ radius (amplitude) $A = 30$. Center 35 m up ⇒ midline $D = 35$. Period 4 min ⇒ $B = 2\pi/4 = \pi/2$. Boarding at the LOWEST point at $t = 0$ means the function starts at its minimum, so use $-\cos$ (which equals -1 at $t = 0$). Final model: $h(t) = -30 \cos(\pi t/2) + 35$.

19

3B · TOPIC 3.10 TRIG EQUATIONS

ANSWER E

Find *all* solutions to $2 \cdot \sin(x) - 1 = 0$ on the interval $[0, 2\pi)$.

A $x = \pi/6$ only

B $x = \pi/3$ only

C $x = \pi/3$ and $x = 2\pi/3$

D $x = \pi/4$ and $x = 3\pi/4$

E $x = \pi/6$ and $x = 5\pi/6$

F $x = \pi/6$ and $x = 7\pi/6$

WORKED SOLUTION

$\sin(x) = 1/2$. Reference angle: $\pi/6$. Sine is positive in QI and QII. QI solution: $x = \pi/6$. QII solution: $x = \pi - \pi/6 = 5\pi/6$. On $[0, 2\pi)$: $x = \pi/6$ and $x = 5\pi/6$.

20 3C · TOPIC 3.14 🔥 POLAR GRAPHS · 6/10 WRONG ANSWER A

Which polar equation produces a **four-petaled rose**?

A $r = 3\cos(2\theta)$

B $r = 4\sin(\theta)$

C $r = 2 + \cos(\theta)$

D $r = 2\cos(3\theta)$

E $r = 1 + 2\cos(\theta)$

F $r = 4$

WORKED SOLUTION

POLAR QUIZ · 6/10 WRONG — REBUILD. For $r = a\cdot\cos(n\theta)$: if n is even, the rose has $2n$ petals; if n is odd, n petals. A four-petaled rose with petals along

the axes ⇒ n = 2 (so 2·2 = 4 petals). Petal length = |a| = 3. ⇒ r = 3 cos(2θ).

21 3C · TOPIC 3.12 DOUBLE-ANGLE ANSWER C

Which expression is equivalent to $\sin(2\theta)$ for all real θ?

A $\sin(\theta) + \cos(\theta)$

B $2\sin(\theta)$

C $2\sin(\theta) \cdot \cos(\theta)$

D $1 - 2\sin^2(\theta)$

E $\cos(\theta) \cdot \sin(\theta)$

F $\sin^2(\theta) - \cos^2(\theta)$

WORKED SOLUTION

Double-angle identity for sine: $\sin(2\theta) = 2\sin\theta\cos\theta$. (Compare: $\cos(2\theta) = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1$ — a different identity.)

UNIT 04 Parametric / Implicit / Conics

22 4A · TOPIC 4.1 FIND PARAMETER ANSWER A

A parametric curve is defined by $x(t) = t^2 - 1$ and $y(t) = 2t + 3$. For which value of t is the point on the curve equal to (3, 7)?

A $t = 2$

B $t = 1$

C $t = -2$

D $t = 3$

E Both $t = 2$ and $t = -2$

F $t = 4$

WORKED SOLUTION

Set the parametric coordinates equal to the target point: $x(t) = t^2 - 1 = 3 \Rightarrow t^2 = 4 \Rightarrow t = \pm 2$. **You must check y for each candidate.** $y(2) = 2(2) + 3 = 7$ ✓ $y(-2) = 2(-2) + 3 = -1$ ✗. Only $t = 2$ works. **TRAP:** option E ("both") is the common mistake from solving x alone.

23

4A · TOPIC 4.3 RATES OF CHANGE

ANSWER C

A parametric curve has $x(t) = 3t^2$ and $y(t) = t^3$. What is the average rate of change of y with respect to x on the interval $t \in [1, 2]$?

A 3

B $\frac{7}{3}$

C $\frac{7}{9}$

D 9

E $\frac{9}{7}$

F $\frac{1}{3}$

WORKED SOLUTION

Average rate of change of y w.r.t. x on a parametric curve = $\Delta y / \Delta x$ using the endpoint values. $y(1) = 1^3 = 1$, $y(2) = 2^3 = 8$ ⇒ $\Delta y = 7$. $x(1) = 3 \cdot 1^2 = 3$, $x(2) = 3 \cdot 2^2 = 12$ ⇒ $\Delta x = 9$. AROC = 7 / 9. COMMON ERROR: flipping to 9/7 (option E).

24

4B · TOPIC 4.4 PARAMETRIC CIRCLE

ANSWER E

Which set of parametric equations describes a complete **circle** of radius 5 centered at (2, −3)?

A $x = 5 + 2\cos(t)$, $y = 5 - 3\sin(t)$, $0 \le t < 2\pi$

B $x = 2\cos(t)$, $y = -3\sin(t)$, $0 \le t < 2\pi$

C $x = 2 + 5t$, $y = -3 + 5t$, $0 \le t < 2\pi$

D $x = 5\cos(t) - 2$, $y = 5\sin(t) + 3$, $0 \le t < 2\pi$

E $x = 2 + 5\cos(t)$, $y = -3 + 5\sin(t)$, $0 \le t < 2\pi$

F $x = 2 + 5\cos(t)$, $y = 3 + 5\sin(t)$, $0 \le t < 2\pi$

WORKED SOLUTION

Standard parametric form for a circle of radius r centered at (h, k): $x = h + r\cos(t)$, $y = k + r\sin(t)$. Center (2, −3), radius 5 ⇒ $x = 2 + 5\cos(t)$, $y = -3 + 5\sin(t)$, $0 \le t < 2\pi$. TRAP: option F has y = +3 instead of −3 (sign error on the center).

25

4B · TOPIC 4.5 IMPLICIT · INTERSECTION

ANSWER F

How many points of intersection are there between the implicit relation $x^2 + y^2 = 25$ and the line $y = x + 1$?

A 0

B 1

C 3

D 4

E Infinitely many

F 2

WORKED SOLUTION

Two ellipses (or one ellipse and a related conic) generically intersect at **2 points** when one passes through the interior of the other on opposite sides. Sketching the two graphs and counting crossings confirms 2 intersection points.

26 4C · TOPIC 4.6 ELLIPSE STANDARD FORM ANSWER B

An ellipse is centered at (1, −2). Its major axis is horizontal with total length 10, and its minor axis is vertical with total length 6. Which equation represents this ellipse in standard form?

A $(x + 1)^2/25 + (y - 2)^2/9 = 1$

B $(x - 1)^2/25 + (y + 2)^2/9 = 1$

C $(x - 1)^2/100 + (y + 2)^2/36 = 1$

D $(x - 1)^2/9 + (y + 2)^2/25 = 1$

E $(x - 1)^2/10 + (y + 2)^2/6 = 1$

F $(x - 1)^2/5 + (y + 2)^2/3 = 1$

WORKED SOLUTION

Standard ellipse form: $(x - h)^2 / a^2 + (y - k)^2 / b^2 = 1$. Center (1, −2), horizontal semi-axis 5, vertical semi-axis 3. ⇒ $(x - 1)^2 / 25 + (y + 2)^2 / 9 = 1$. **TRAP:** options using 5 and 3 (not 25 and 9) skip the squaring step (option F).

27

4C · TOPIC 4.6 🔥 HYPERBOLA · Q20 MISS

ANSWER F

For the hyperbola $(x - 2)^2/16 - (y + 1)^2/9 = 1$, what are the equations of the two asymptotes?

A $y = \pm(4/3)(x - 2) - 1$

B $y = \pm(3/4)(x + 2) + 1$

C $y = \pm(16/9)(x - 2) - 1$

D $y = \pm 3(x - 2) - 1$

E $y = (3/4)\,x$ and $y = -(3/4)\,x$

F $y = \pm(3/4)(x - 2) - 1$

WORKED SOLUTION

Hyperbola asymptotes: for a horizontal-opening hyperbola centered at (h, k) with semi-axes a, b: $y - k = \pm(b/a)(x - h)$. Center (2, −1), a = 4 (horizontal), b = 3 (vertical) ⇒ $y = \pm(3/4)(x - 2) - 1$.

28

4C · TOPIC 4.7 PARAMETRIC CONIC

ANSWER B

The parametric curve $x = 4\cos(t)$, $y = 3\sin(t)$, $0 \le t < 2\pi$, traces:

A A circle of radius 4

B An ellipse with semi-major axis 4 (horizontal) and semi-minor axis 3 (vertical)

C A circle of radius 3

D A hyperbola

E A line through the origin

F A parabola

WORKED SOLUTION

Eliminate the parameter using $\cos^2 t + \sin^2 t = 1$: $x/4 = \cos t$, $y/3 = \sin t$ ⇒ $(x/4)^2 + (y/3)^2 = 1$ ⇒ $x^2/16 + y^2/9 = 1$. That's an **ellipse** with horizontal semi-axis $a = 4$ and vertical semi-axis $b = 3$. **TELLTALE:** different coefficients on cos and sin ⇒ ellipse (equal coefficients ⇒ circle).

FREE RESPONSE

Q29–Q30 · Worked Solutions

29

3B · TOPIC 3.6 / 3.10 SINUSOIDAL MODELING · 4 SUB-PARTS

Tidal harbor: water level is sinusoidal. **Maximum 22 ft at t = 6 hr**, **minimum 6 ft at t = 18 hr** (t in hours after midnight). Find the model and evaluate.

(a) Find the amplitude A and the midline D.

ANSWER $A = 8$, $D = 14$

WORK

Amplitude is half the vertical range: $A = (\max - \min) / 2 = (22 - 6) / 2 = 8$. Midline is the average of max and min: $D = (\max + \min) / 2 = (22 + 6) / 2 = 14$.

(b) Find the period T (in hours) and the value of B.

ANSWER | Period = 24 hours, B = π/12

WORK

From a maximum at t = 6 to a minimum at t = 18 is half of a full period: 12 hours = T/2 ⇒ T = 24 hours. Then B = 2π / T = 2π / 24 = π/12.

(c) Write the full sinusoidal equation h(t) using cosine.

ANSWER | h(t) = 8 cos((π/12)(t − 6)) + 14

WORK

Cosine is at its maximum when its argument is 0. We want the max at t = 6, so the phase shift is C = 6: cos((π/12)(t − 6)) peaks at t = 6. Multiply by amplitude 8 and add midline 14: h(t) = 8 cos((π/12)(t − 6)) + 14.

(d) Evaluate h(12).

ANSWER | h(12) = 14 ft (the midline)

WORK

h(12) = 8 cos((π/12)(12 − 6)) + 14 = 8 cos(π/2) + 14. cos(π/2) = 0, so h(12) = 8(0) + 14 = 14 ft. Geometrically, t = 12 is a quarter-period after the maximum at t = 6, so the function is crossing the midline going downward.

30

4C · TOPIC 4.7 / 4.10 HYPERBOLA → PARAMETRIC · 3 SUB-PARTS

Hyperbola: $(x - 1)^2 / 9 - (y + 2)^2 / 16 = 1$. Identify key features and a parametric form.

(a) Identify the center and the two vertices.

ANSWER | Center: (1, −2). Vertices: (4, −2) and (−2, −2)

WORK

Standard form: $(x - h)^2/a^2 - (y - k)^2/b^2 = 1$, horizontal-opening. Read off $h = 1$, $k = -2$, $a^2 = 9 \Rightarrow a = 3$, $b^2 = 16 \Rightarrow b = 4$. Center = (h, k) = (1, −2). Vertices on the horizontal axis through the center, distance a = 3 left and right: (1 + 3, −2) = (4, −2) and (1 − 3, −2) = (−2, −2).

(b) Write the equations of the two asymptotes (slope-intercept form).

ANSWER | $y = (4/3)(x - 1) - 2$ and $y = -(4/3)(x - 1) - 2$

WORK

Asymptote formula for horizontal hyperbola: $y - k = \pm(b/a)(x - h)$. With $h = 1$, $k = -2$, $b/a = 4/3$: $y = \pm(4/3)(x - 1) - 2$. Either ordering accepted.

(c) Give a parametric form using sec(t) and tan(t), then evaluate at t = 0.

ANSWER

$x(t) = 1 + 3\sec(t)$, $y(t) = -2 + 4\tan(t)$. At t = 0: (4, −2)

WORK

Standard parametric form for a horizontal hyperbola: $x = h + a\sec(t)$, $y = k + b\tan(t)$ (uses the identity $\sec^2 t - \tan^2 t = 1$, which mirrors $x^2/a^2 - y^2/b^2 = 1$). Plug in: $x(t) = 1 + 3\sec(t)$, $y(t) = -2 + 4\tan(t)$. At t = 0: $\sec(0) = 1$, $\tan(0) = 0 \Rightarrow x = 1 + 3 = 4$, $y = -2 + 0 = -2 \Rightarrow (4, -2)$ (which is the right vertex — the parametric form starts on a vertex).

Stratosyn · 2026 Built for Room 501 · ALA Queen Creek